

A Log-Skill and Two-Dimensional Paradigm-Switching Model of Output Growth: Gain, Drift, Timing, and Real Options

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Abstract

This paper develops a stylized model of long-horizon output growth under learning, bottlenecks, and paradigm switches. The revised formulation separates three ideas that are often conflated: within-paradigm skill, paradigm gain, and paradigm drift. We first review canonical models that already capture parts of this structure, including learning curves, multiplicative production functions, diffusion and adoption dynamics, switching costs, and real-options timing. We then provide a compact model in which output speed is $V = e^x R$, where x is a dimensionless log-skill state and R is the cumulative multiplier from previous paradigm gains. A new paradigm is represented by a two-dimensional vector $r = (g, \delta)$: gain $g > 0$ measures the intrinsic multiplier of the new paradigm, while drift $\delta \geq 0$ measures distance from the old paradigm and therefore loss of skill transfer. A switch maps $x^+ = h(\delta)x^-$ and $R^+ = R^- g$, where $h(\delta) = e^{-\alpha\delta}$. The instantaneous switching ratio is

$$\frac{V^+}{V^-} = g \exp\{(h(\delta) - 1)x^-\},$$

yielding a no-pain log-skill threshold $x^*(g, \delta) = \ln g / (1 - h(\delta))$. The log-state formulation addresses the arbitrary unit-scaling problem of the earlier K^h representation by anchoring skill to a novice baseline and modeling transfer in log-skill space. We also derive a general path-dependent total-output formula, a corrected finite-horizon switching objective, a repeated-switching interpretation that does not suppress drift, and a real-options extension for staged validation when gain and drift are uncertain. The real-options section distinguishes perfect information from noisy pilots, explicitly incorporates information quality and niche-decay costs, and shows how the same threshold logic can be used with integrated finite-horizon payoffs. The framework explains why experts and incumbents may be locally rational in delaying adoption, why newer adopters may adopt high-gain tools faster, why incumbents should sometimes self-disrupt immediately when gain is verified and niche advantage decays, and why unvalidated transformation can be costly when gain is low and drift is high.

Keywords: learning curves; paradigm switching; innovation; organizational inertia; switching costs; first-mover advantage; real options; dynamic capabilities; output growth; human capital.

JEL-style tags: O31, O32, D83, L25, M10, G31.

1 Introduction

Output growth in research, entrepreneurship, technology, and organizational life is not simply a function of elapsed time. Production can create reusable assets: skill, schemas, templates, routines, code, reputation, network access, and judgment. Yet real production systems are not smooth. They encounter bottlenecks. Some bottlenecks are solved by incremental practice; others require a change of paradigm. Such changes often have a characteristic shape: short-run disruption and skill loss, followed by possible long-run acceleration.

The central question of this paper is how to formalize that tradeoff. An agent may delay adopting a new method not because the agent is irrational, but because the agent has high legacy skill. A new method may be superior in the long run but still reduce short-run output. Conversely, a new method may be fashionable but inferior, in which case it both reduces multiplier value and wastes legacy skill. The model therefore rejects two symmetrical errors: naive conservatism and naive innovation.

The first draft of this model used a direct update $K^+ = K^{h(\delta)}$. A rigorous revision requires care: if K is interpreted as a raw skill quantity with arbitrary units, exponentiation is not scale invariant. The corrected model therefore uses a dimensionless productivity multiple κ and its log state

$$x = \ln \kappa.$$

Paradigm drift compresses this log-skill state, not an arbitrary physical unit of skill. The core speed equation becomes

$$V = e^x R.$$

A paradigm switch is represented by a two-dimensional vector

$$r = (g, \delta),$$

where $g > 0$ is the paradigm gain and $\delta \geq 0$ is the paradigm drift. The transition is

$$x^+ = h(\delta)x^-, \quad R^+ = R^- g,$$

with $h(\delta) = e^{-\alpha\delta}$. Thus

$$\frac{V^+}{V^-} = g e^{(h(\delta)-1)x^-}.$$

This compact expression is the core of the paper. Gain determines whether a new paradigm is intrinsically better; drift determines how much legacy log-skill survives.

2 Existing Building Blocks

The model is not intended to replace existing literatures. It packages several familiar ideas into a single output-growth framework and adds one central distinction: paradigm gain and paradigm drift are separate objects.

2.1 Learning curves and learning by doing

The closest existing production-side ancestor is the learning curve. Wright’s law is often written as

$$C(Q) = C_0 Q^{-b}, \quad b > 0, \quad (1)$$

where $C(Q)$ is unit cost at cumulative output Q . Equivalently,

$$\ln C(Q) = \ln C_0 - b \ln Q. \quad (2)$$

This is a log-log relation: cumulative output maps into proportional improvement. Arrow’s learning-by-doing argument similarly treats production experience as a source of productivity. These models motivate the use of relative, logarithmic states rather than arbitrary absolute skill units.

2.2 Multiplicative production functions

Cobb-Douglas production takes the form

$$Y = AK_c^\alpha L^\beta, \quad (3)$$

where A is total factor productivity, K_c is capital, and L is labor. Our notation K in the first draft was not physical capital and is avoided here. The useful connection is multiplicativity: output is often modeled as a product of state terms and productivity multipliers. In our model, e^x is within-paradigm productivity and R is a cumulative paradigm multiplier.

2.3 Diffusion, adoption, and niche timing

Technology adoption and market capture are often modeled with diffusion. The Bass model, for instance, can be written as

$$\frac{dN}{dt} = \left(p + q \frac{N}{M} \right) (M - N), \quad (4)$$

where $N(t)$ is cumulative adopters, M is market potential, p captures external innovation effects, and q captures imitation. We do not reproduce the full Bass model, but we borrow the idea that market position and adoption windows are time dependent. We represent the available niche or ecosystem advantage after delay τ by $S(\tau)$, usually with $S'(\tau) < 0$.

2.4 Switching costs, real options, and exploration-exploitation

Switching-cost models emphasize that changing suppliers, technologies, routines, or standards is costly even when the target state is better. Real-options theory adds that waiting has value when investment payoffs are uncertain, but waiting can also destroy first-mover advantage. Exploration-exploitation models describe the tension between refining current routines and searching for better ones. Our model operationalizes these ideas using two variables: gain g and drift δ .

3 Ratio-Based Learning and the Log State

Let $O(t)$ denote cumulative effective output and $V(t) = dO(t)/dt$ denote instantaneous output speed. A simple intuition is that output speed improves with accumulated output, because past output encodes skill and reusable assets. But the more precise learning intuition is often ratio based: a doubling from 10 to 20 units may be comparable to a doubling from 100 to 200 units.

Assumption 1 (Ratio-based speed increments). *For $O_a, O_b > 0$, the speed increment from O_a to O_b depends only on the ratio O_b/O_a :*

$$V(O_b) - V(O_a) = F\left(\frac{O_b}{O_a}\right). \quad (5)$$

Proposition 1 (Logarithmic relation). *If F is continuous and speed increments over successive ratio changes are well defined, then*

$$F(x) = k \log_b x \quad (6)$$

for some k and base $b > 1$. Hence

$$V(O) = V(O_0) + k \log_b \left(\frac{O}{O_0}\right). \quad (7)$$

Proof. Let $x = O_b/O_a$ and $y = O_c/O_b$. Then $O_c/O_a = xy$. The speed increment from O_a to O_c is the sum of the increments from O_a to O_b and from O_b to O_c , so

$$F(xy) = F(x) + F(y). \quad (8)$$

The continuous solutions of this multiplicative Cauchy equation are logarithmic. □

This result is not the final production law of the paper. Its role is to justify the log-state representation: relative improvement naturally suggests modeling skill on a logarithmic scale. The remainder of the paper uses $x = \ln \kappa$ as the state variable.

4 Core Model: Log Skill, Gain, and Drift

Definition 1 (Dimensionless within-paradigm productivity). *Let $\kappa \geq 1$ denote current within-paradigm productivity relative to a novice baseline. It has no physical unit. Define the log-skill state*

$$x = \ln \kappa. \quad (9)$$

Definition 2 (Cumulative paradigm multiplier). *Let $R > 0$ denote the cumulative multiplier from previous paradigm gains. Instantaneous output speed is*

$$V = e^x R. \quad (10)$$

Definition 3 (Paradigm vector). *A new paradigm is represented by*

$$r = (g, \delta), \quad (11)$$

where $g > 0$ is paradigm gain and $\delta \geq 0$ is paradigm drift.

Gain and drift are distinct. A paradigm can be high-gain and low-drift, high-gain and high-drift, low-gain and low-drift, or low-gain and high-drift. This distinction is the main conceptual contribution of the model.

Assumption 2 (Skill-transfer function). *Legacy log-skill transfers according to*

$$x^+ = h(\delta)x^-. \quad (12)$$

The function $h(\delta)$ satisfies

$$h(0) = 1, \quad h'(\delta) < 0, \quad \lim_{\delta \rightarrow \infty} h(\delta) = 0. \quad (13)$$

The default form is

$$h(\delta) = e^{-\alpha\delta}, \quad \alpha > 0. \quad (14)$$

Unless otherwise stated, $\alpha = 1$.

The cumulative multiplier updates as

$$R^+ = R^- g. \quad (15)$$

Thus the post-switch speed is

$$V^+ = e^{h(\delta)x^-} R^- g. \quad (16)$$

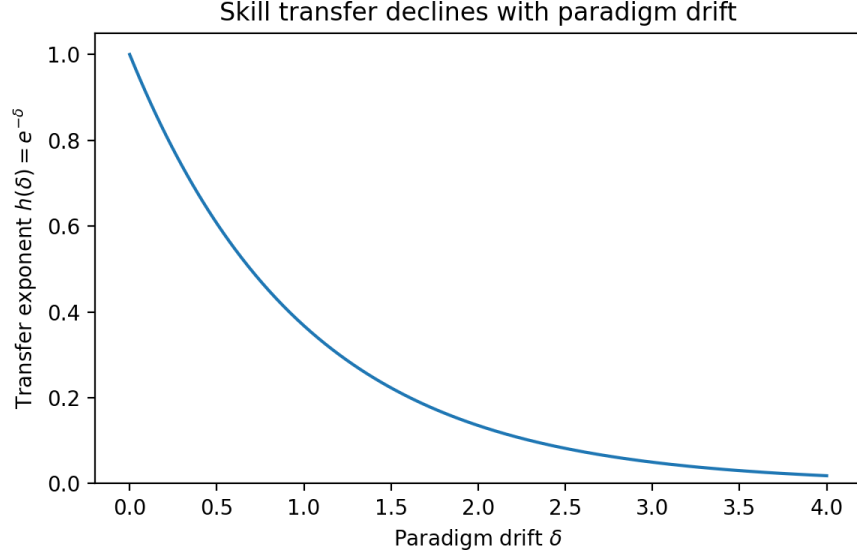


Figure 1: The transfer exponent $h(\delta) = e^{-\delta}$. Drift reduces the share of legacy log-skill that survives into the new paradigm.

5 Core Results

Proposition 2 (Switching ratio). *If pre-switch speed is $V^- = e^{x^-} R^-$ and the agent switches to $r = (g, \delta)$, then*

$$\frac{V^+}{V^-} = g \exp\{(h(\delta) - 1)x^-\}. \quad (17)$$

Proof. By the transition rules,

$$V^+ = e^{h(\delta)x^-} R^- g. \quad (18)$$

Dividing by $V^- = e^{x^-} R^-$ gives the result. $\square$

Proposition 3 (No-pain log-skill threshold). *Suppose $g > 1$ and $h(\delta) < 1$. The switch is instantaneously non-decreasing in speed if and only if*

$$x^- \leq x^*(g, \delta), \quad x^*(g, \delta) = \frac{\ln g}{1 - h(\delta)}. \quad (19)$$

Equivalently, in dimensionless productivity multiples,

$$\kappa^- \leq \kappa^*(g, \delta), \quad \kappa^*(g, \delta) = g^{1/(1-h(\delta))}. \quad (20)$$

Proof. The switch is non-decreasing when

$$ge^{(h-1)x} \geq 1. \quad (21)$$

Taking logs,

$$\ln g + (h - 1)x \geq 0. \quad (22)$$

Since $h < 1$, this is equivalent to $x \leq \ln g / (1 - h)$. $\square$

Corollary 1 (High legacy skill creates local conservatism). *For fixed $g > 1$ and $\delta > 0$, the switching ratio is strictly decreasing in x .*

Proof. The derivative of $ge^{(h-1)x}$ with respect to x is $g(h-1)e^{(h-1)x} < 0$. $\square$

Corollary 2 (Bad change is doubly costly). *If $0 < g < 1$, $x \geq 0$, and $\delta > 0$, then $V^+ / V^- < 1$.*

Proof. For $0 < g < 1$ and $h < 1$, $e^{(h-1)x} \leq 1$ when $x \geq 0$. Thus $ge^{(h-1)x} < 1$. $\square$

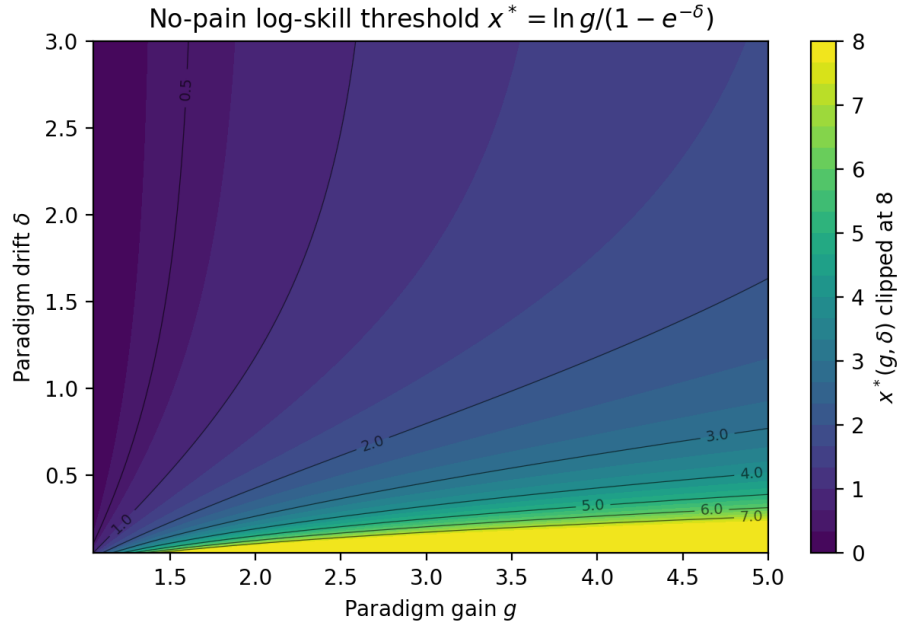


Figure 2: No-pain log-skill threshold $x^*(g, \delta) = \ln g / (1 - e^{-\delta})$. Larger gain raises the threshold; larger drift lowers transfer and makes switching painful at lower legacy log-skill.

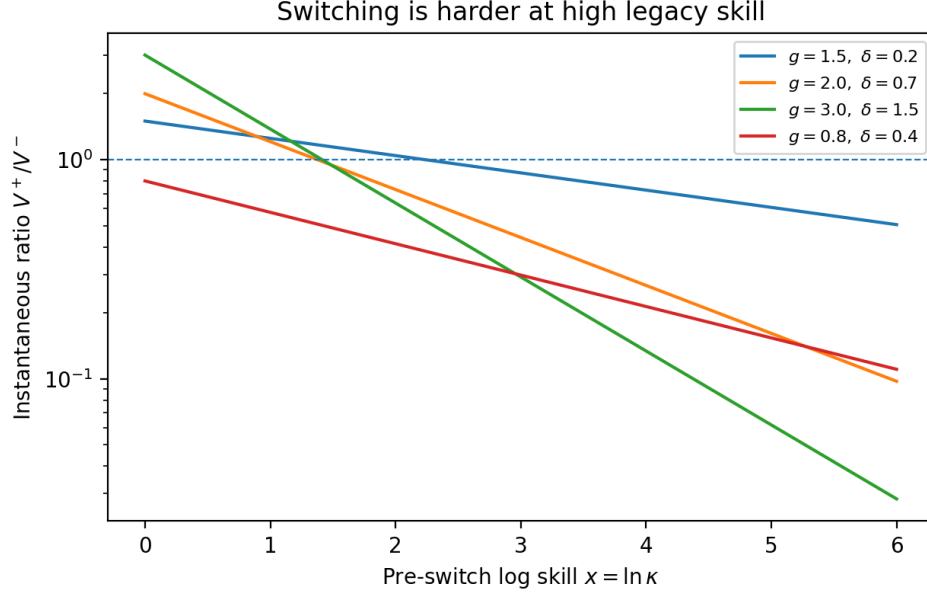


Figure 3: Instantaneous switching ratio as a function of pre-switch log skill. Even high-gain switches become initially painful at high legacy skill when drift is substantial.

6 General Path-Dependent Output Formula

Let paradigm switches occur at

$$0 < t_1 < t_2 < \dots < t_n < T. \quad (23)$$

Define

$$t_0 = 0, \quad t_{n+1} = T, \quad \Delta_i = t_{i+1} - t_i. \quad (24)$$

At switch i , the paradigm vector is $r_i = (g_i, \delta_i)$ and $h_i = h(\delta_i)$.

Let $\Phi_s(x)$ denote within-paradigm log-skill evolution over duration s starting from log-skill x . This avoids committing the general model to a single learning law. It only requires $\Phi_0(x) = x$ and non-decreasing skill in s .

Let x_i^+ be log-skill immediately after switch i , with initial state x_0^+ at $t_0 = 0$. For $i \geq 1$,

$$x_i^+ = h_i \Phi_{\Delta_{i-1}}(x_{i-1}^+). \quad (25)$$

Let

$$G_i = \prod_{j=1}^i g_j, \quad G_0 = 1. \quad (26)$$

Then on interval $[t_i, t_{i+1})$,

$$V(t) = G_i \exp\{\Phi_{t-t_i}(x_i^+)\}. \quad (27)$$

Therefore total output is

$$O(T) = \sum_{i=0}^n G_i \int_0^{\Delta_i} \exp\{\Phi_s(x_i^+)\} ds. \quad (28)$$

This is the general path-dependent form. Gains compound multiplicatively; drift repeatedly compresses the log-skill state.

6.1 Linear-productivity benchmark

For continuity with the first draft, consider the special benchmark in which dimensionless productivity $\kappa = e^x$ grows linearly within a paradigm:

$$\kappa(t_i + s) = \kappa_i + s. \quad (29)$$

Then

$$O(T) = \sum_{i=0}^n G_i \left(\kappa_i \Delta_i + \frac{1}{2} \Delta_i^2 \right), \quad (30)$$

with

$$\kappa_i = (\kappa_{i-1} + \Delta_{i-1})^{h_i}, \quad \kappa_0 = 1. \quad (31)$$

This formula should be interpreted as a benchmark, not as a universal law.

6.2 Explicit expansion under the linear-productivity benchmark

The first terms are

$$O(T) = \Delta_0 + \frac{1}{2} \Delta_0^2 \quad (32)$$

$$+ g_1 \left[(1 + \Delta_0)^{h_1} \Delta_1 + \frac{1}{2} \Delta_1^2 \right] \quad (33)$$

$$+ g_1 g_2 \left[\left((1 + \Delta_0)^{h_1} + \Delta_1 \right)^{h_2} \Delta_2 + \frac{1}{2} \Delta_2^2 \right] \quad (34)$$

$$+ g_1 g_2 g_3 \left[\left(\left((1 + \Delta_0)^{h_1} + \Delta_1 \right)^{h_2} + \Delta_2 \right)^{h_3} \Delta_3 + \frac{1}{2} \Delta_3^2 \right] + \dots \quad (35)$$

7 Corrected Timing: From Myopic Speed to Finite-Horizon Output

An organization facing a paradigm shift must decide when to switch. A useful but limited calculation is the myopic post-switch speed objective. Suppose the firm waits τ , old dimensionless productivity grows from κ_0 to $\kappa_0 + \tau$ in the linear-productivity benchmark, and niche advantage decays as

$$S(\tau) = e^{-\beta\tau}. \quad (36)$$

Post-switch speed is proportional to

$$(\kappa_0 + \tau)^h e^{-\beta\tau}. \quad (37)$$

Maximizing this expression gives

$$\tau_{\text{myopic}}^* = \max \left(0, \frac{h}{\beta} - \kappa_0 \right). \quad (38)$$

This optimizes instantaneous post-switch speed, not total output.

The finite-horizon objective is different. Suppose there is one possible switch before horizon T . If the organization switches at τ , total output is

$$J(\tau) = R \left(\kappa_0 \tau + \frac{1}{2} \tau^2 \right) + c g e^{-\beta\tau} R \left[(\kappa_0 + \tau)^h (T - \tau) + \frac{1}{2} (T - \tau)^2 \right]. \quad (39)$$

Let

$$B(\tau) = (\kappa_0 + \tau)^h (T - \tau) + \frac{1}{2} (T - \tau)^2. \quad (40)$$

Then

$$J'(\tau) = R(\kappa_0 + \tau) + c g R e^{-\beta\tau} (B'(\tau) - \beta B(\tau)), \quad (41)$$

where

$$B'(\tau) = h(\kappa_0 + \tau)^{h-1} (T - \tau) - (\kappa_0 + \tau)^h - (T - \tau). \quad (42)$$

The true finite-horizon optimum is a boundary solution or an interior root of $J'(\tau) = 0$. This formulation makes explicit the opportunity cost of waiting: delay preserves old-paradigm skill for a while, but it postpones the gain g and erodes the niche coefficient $S(\tau)$ over the remaining horizon.

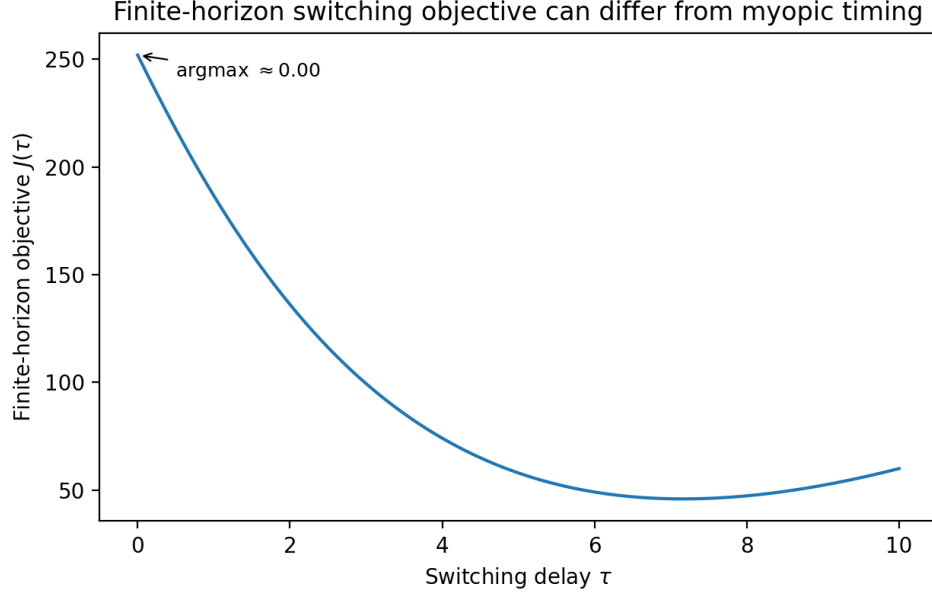


Figure 4: A finite-horizon timing objective. Optimizing total output can differ from optimizing post-switch speed. Under the displayed parameterization, the maximum is a boundary solution at immediate switching ($\tau = 0$).

8 Repeated Switching with Drift

A high-frequency paradigm-switching path should not be written as $(t + 1)e^{at}$ unless drift is absent or negligible. Suppose switches occur at intervals of length Δ and have common drift exponent $h \in (0, 1)$. Under the linear-productivity benchmark,

$$\kappa_{n+1} = (\kappa_n + \Delta)^h. \quad (43)$$

This map may have a finite fixed point κ_s^* satisfying

$$\kappa_s^* = (\kappa_s^* + \Delta)^h. \quad (44)$$

Thus repeated high-drift switching can prevent within-paradigm skill from accumulating without bound. The exponential component, if present, comes from cumulative gains:

$$G_n = \prod_{i=1}^n g_i, \quad \frac{1}{n} \ln G_n \rightarrow E[\ln g_i] \quad (45)$$

under standard regularity assumptions. With switch frequency λ , the gain multiplier grows approximately as

$$G(t) \approx \exp\{\lambda E[\ln g]t\}. \quad (46)$$

Therefore repeated switching is exponential-dominant only if average log gain is positive. Drift may keep the skill term bounded, but it does not eliminate exponential growth in $G(t)$ when $E[\ln g] > 0$.

9 Organizational Extension

Organizations face additional frictions. Let $c \in (0, 1]$ capture coordination efficiency: bureaucracy, incentive conflict, communication overhead, training cost, and internal politics. Let $S(\tau) > 0$ denote niche or ecosystem advantage after delay τ from the emergence of the new paradigm. Usually $S'(\tau) < 0$.

Organizational post-switch speed is

$$V_{\text{org}}^+ = cS(\tau)e^{h(\delta)x^-}R^-g. \quad (47)$$

The ratio is

$$\frac{V_{\text{org}}^+}{V_{\text{org}}^-} = cS(\tau)ge^{(h(\delta)-1)x^-}. \quad (48)$$

If $cS(\tau)g > 1$, the organizational no-pain log-skill threshold is

$$x_{\text{org}}^* = \frac{\ln[cS(\tau)g]}{1 - h(\delta)}. \quad (49)$$

This expression clarifies an incumbent paradox. A large incumbent may have enormous legacy skill and assets. Even after drift compression, e^{hx} may remain larger than a startup's entire productivity base. If g is verified and $S(\tau)$ decays quickly, immediate self-disruption can be optimal. But if g is uncertain or low and δ is high, full migration can destroy value.

10 Staged Validation as a Real-Options Problem

The previous sections often treat (g, δ) as known. In practice, a decision maker rarely knows whether a new paradigm has sufficiently high gain, sufficiently low drift, or sufficiently durable niche advantage. This section gives a minimal real-options characterization rather than only a schematic dynamic program. The results should be read as staged-validation criteria: pilots, prototypes, sandbox teams, and limited deployments are valuable when they improve the decision between staying and switching enough to compensate for direct cost, imperfect information, and delay.

10.1 A two-state gain-drift model

Let current log-skill be x , cumulative multiplier be R , and remaining horizon be H . Let

$$\Lambda = cS(\tau) \quad (50)$$

collect coordination efficiency and currently available niche advantage. For an individual decision with no organizational friction, set $\Lambda = 1$.

Instead of making only gain uncertain, suppose the new paradigm has two possible gain-drift states:

$$\theta = \begin{cases} \theta_H = (g_H, \delta_H), & \text{with probability } p, \\ \theta_L = (g_L, \delta_L), & \text{with probability } 1 - p, \end{cases} \quad (51)$$

with $h_j = h(\delta_j)$ for $j \in \{H, L\}$. The labels H and L refer to the high-payoff and low-payoff switching states, not necessarily to gain alone. If the agent stays in the old paradigm for the horizon, the flat-speed benchmark payoff is

$$W = e^x RH. \quad (52)$$

If the agent switches and state $j \in \{H, L\}$ obtains, the payoff is

$$S_j = \Lambda g_j e^{h_j x} RH, \quad (53)$$

and assume $S_H > S_L$. The no-experiment value is therefore

$$V_0(p) = \max\{pS_H + (1-p)S_L, W\}. \quad (54)$$

Proposition 4 (Belief threshold for switching under gain-drift uncertainty). *Assume $S_H > S_L$. The agent switches without experimentation if and only if*

$$p \geq p^* = \frac{W - S_L}{S_H - S_L}. \quad (55)$$

When $p^ \leq 0$, switching is optimal for all beliefs; when $p^* \geq 1$, staying is optimal for all beliefs.*

Proof. Switching is preferred when

$$pS_H + (1-p)S_L \geq W. \quad (56)$$

Rearranging yields

$$p(S_H - S_L) \geq W - S_L, \quad (57)$$

which gives the threshold. $\square$

If only gain is uncertain and drift is fixed, $h_H = h_L = h$, this reduces to the simpler structural expression

$$p^* = \frac{e^{(1-h)x}/\Lambda - g_L}{g_H - g_L}. \quad (58)$$

In that special case, higher legacy log-skill x , higher drift δ through lower h , and lower Λ all raise the belief required for switching. With uncertain drift, the common factor no longer cancels; the threshold still exists, but evidence must identify not only whether the new paradigm is high-gain

but also how much legacy log-skill survives. This is why a pilot should measure both improvement after adoption and transfer loss during retraining.

10.2 Value of a perfectly revealing experiment

Now suppose the agent can run a one-shot experiment at cost C_e . The experiment perfectly reveals whether the state is H or L , after which the agent chooses optimally between switching and staying. The experiment value is

$$V_E(p) = -C_e + p \max\{S_H, W\} + (1 - p) \max\{S_L, W\}. \quad (59)$$

Proposition 5 (Value of perfect staged validation). *A perfectly revealing experiment is optimal if and only if*

$$p \max\{S_H, W\} + (1 - p) \max\{S_L, W\} - \max\{pS_H + (1 - p)S_L, W\} > C_e. \quad (60)$$

In the nontrivial case $S_H > W > S_L$, the gross value of information is

$$I(p) = \begin{cases} p(S_H - W), & p < p^*, \\ (1 - p)(W - S_L), & p \geq p^*, \end{cases} \quad (61)$$

where $p^ = (W - S_L)/(S_H - S_L)$. The experiment is worthwhile exactly when $I(p) > C_e$.*

Proof. The first inequality compares the payoff of experimenting to the best non-experiment action. In the nontrivial case $S_H > W > S_L$, a favorable signal leads to switching and an unfavorable signal leads to staying. If $p < p^*$, the no-experiment action is to stay, so the gross information value is

$$pS_H + (1 - p)W - W = p(S_H - W). \quad (62)$$

If $p \geq p^*$, the no-experiment action is to switch, so the gross information value is

$$pS_H + (1 - p)W - [pS_H + (1 - p)S_L] = (1 - p)(W - S_L). \quad (63)$$

Subtracting C_e gives the net value condition. □

This proposition is a value-of-perfect-information benchmark. It is useful because it gives an upper bound on the value of experimentation and shows why information is most valuable near the switching boundary. It should not be interpreted as the value of a realistic pilot unless the pilot is close to fully revealing and imposes no material delay.

10.3 Noisy staged validation with niche-decay cost

A more realistic pilot produces an imperfect signal and consumes time. Let the experiment generate a binary signal $y \in \{H, L\}$ with accuracy $q \in [1/2, 1]$:

$$\Pr(y = H \mid H) = q, \quad \Pr(y = L \mid L) = q. \quad (64)$$

The signal probabilities are

$$\pi_H = \Pr(y = H) = qp + (1 - q)(1 - p), \quad (65)$$

$$\pi_L = \Pr(y = L) = (1 - q)p + q(1 - p). \quad (66)$$

The posterior beliefs are

$$p_H = \Pr(H \mid y = H) = \frac{qp}{qp + (1 - q)(1 - p)}, \quad (67)$$

$$p_L = \Pr(H \mid y = L) = \frac{(1 - q)p}{(1 - q)p + q(1 - p)}. \quad (68)$$

Let the experiment take $\varepsilon \geq 0$ units of time. During that delay, the niche coefficient falls from $\Lambda = cS(\tau)$ to

$$\Lambda_\varepsilon = cS(\tau + \varepsilon) = d_\varepsilon \Lambda, \quad d_\varepsilon = \frac{S(\tau + \varepsilon)}{S(\tau)} \leq 1. \quad (69)$$

Under the same flat-speed benchmark, post-experiment payoffs over the remaining horizon are

$$W_\varepsilon = e^x R(H - \varepsilon), \quad (70)$$

$$S_{j,\varepsilon} = \Lambda_\varepsilon g_j e^{h_j x} R(H - \varepsilon), \quad j \in \{H, L\}. \quad (71)$$

Let $\rho \in [0, 1]$ denote the fraction of old-paradigm production preserved during the experiment. Then the payoff of running the noisy experiment is

$$V_E^q(p, \varepsilon) = \rho e^x R\varepsilon - C_e \quad (72)$$

$$+ \pi_H \max\{p_H S_{H,\varepsilon} + (1 - p_H) S_{L,\varepsilon}, W_\varepsilon\} \quad (73)$$

$$+ \pi_L \max\{p_L S_{H,\varepsilon} + (1 - p_L) S_{L,\varepsilon}, W_\varepsilon\}. \quad (74)$$

Proposition 6 (Noisy staged-validation criterion). *The noisy experiment is worthwhile if and only if*

$$V_E^q(p, \varepsilon) > V_0(p). \quad (75)$$

If $q = 1$, $\varepsilon = 0$, and ρ is irrelevant, this criterion reduces to the perfectly revealing benchmark above. If $q = 1/2$, the signal does not change beliefs; absent other benefits, positive direct cost or niche-decay delay makes the experiment weakly worse than choosing immediately.

Proof. After observing a noisy signal, the decision maker updates beliefs by Bayes' rule and then

compares the posterior expected switching payoff with the posterior staying payoff. The displayed expression is the expected value of those signal-contingent optimal decisions, net of direct cost and including production retained during the experiment. Comparing it with $V_0(p)$ gives the criterion. The limiting cases follow directly from the posterior formulas. $\square$

This result connects staged validation to timing. A pilot is not free simply because it is smaller than full migration. It can preserve downside protection by reducing uncertainty, but it also delays commitment and erodes $S(\tau)$. The correct question is therefore not “should we experiment?” in the abstract. It is whether the expected improvement in the stay-versus-switch decision exceeds direct cost plus the lost option value caused by delay.

10.4 Integrated payoffs and dynamic extension

The flat-speed payoffs above are intentionally minimal. They make the real-options logic transparent, but they suppress within-paradigm skill accumulation. The propositions do not require that simplification. They only require three payoff primitives: the value of staying, the value of switching in the high state, and the value of switching in the low state.

For example, using the general skill-evolution map Φ_s , the value of staying over horizon H can be written as

$$W^\Phi = R \int_0^H \exp\{\Phi_s(x)\} ds. \quad (76)$$

If experimentation lasts ε , preserves fraction ρ of old-paradigm production during the pilot, and then switches in state j , the integrated switching payoff can be written as

$$S_{j,\varepsilon}^\Phi = \rho R \int_0^\varepsilon \exp\{\Phi_s(x)\} ds \quad (77)$$

$$+ cS(\tau + \varepsilon)g_j R \int_0^{H-\varepsilon} \exp\{\Phi_s(h_j\Phi_\varepsilon(x))\} ds. \quad (78)$$

Replacing (W, S_H, S_L) by $(W^\Phi, S_{H,\varepsilon}^\Phi, S_{L,\varepsilon}^\Phi)$ preserves the same threshold and information-value logic, while reconnecting the real-options section to the path-dependent output formula in Section 6.

A richer dynamic model lets

$$\theta = (g, \delta, \beta, c) \quad (79)$$

be an uncertain state with belief $b_t(\theta)$. At each date, the agent chooses

$$a_t \in \{\text{stay, switch, experiment}\}. \quad (80)$$

An experiment costs C_e , takes time ε , generates a signal $y \sim p(y \mid \theta)$, and produces a Bayesian update $b_{t+1} = B(b_t, y)$. A schematic dynamic program is

$$V(x, R, b, T) = \max\{V_{\text{stay}}, V_{\text{switch}}, V_{\text{experiment}}\}, \quad (81)$$

where

$$V_{\text{experiment}} = -C_e + E_{y \sim p(\cdot|b)} [V(x', R, b', T - \varepsilon)] . \quad (82)$$

The two-state propositions above are the closed-form finite version of this broader real-options logic.

11 Interpretive Examples

11.1 High-legacy-skill experts and new tools

A high-legacy-skill expert has high x in the old paradigm. Even when $g > 1$, any positive drift makes $h(\delta) < 1$, so the switching ratio declines with x . The expert's claim that the new tool initially feels slower may be locally true. Delayed adoption can arise from real skill-transfer loss rather than ignorance.

11.2 Newer adopters and AI-assisted work

A newer adopter has lower old-paradigm log-skill. For the same high-gain, moderate-drift tool, the switching ratio is higher. This helps explain why people with less legacy investment in a specific workflow may adopt AI-assisted workflows faster than high-legacy-skill experts.

11.3 Cross-disciplinary projects

A domain expert choosing between a familiar frontier problem and a cross-disciplinary design or playtesting project may face comparable potential gain in both directions. But the drift differs. The familiar project has lower drift because core judgment, methods, and literature knowledge transfer. The cross-disciplinary project has higher drift because domain intuitions, user-experience judgment, and prototyping practices transfer less. Comparable gain does not imply comparable near-term productivity.

11.4 Incumbents, hype, and premature transformation

A large organization may pursue a technology with high imagined future gain. If currently verified gain is low while drift is high, full-system migration can erode value: it compresses old log-skill and fails to compensate with gain. The correct policy is often real-options exposure: preserve the option, invest in learning, run experiments, and scale only when evidence about gain, drift, and timing improves.

12 Strategic Implications

1. Do not confuse gain with transferability. A direction can be valuable in the abstract but distant from current competence.
2. High legacy skill can make delayed adoption locally rational. Senior experts and incumbents may delay adoption because the switching ratio declines with legacy log-skill x .

3. Unvalidated innovation can be costly. If $g < 1$, a switch is doubly costly.
4. Verified high gain plus decaying niche advantage justifies speed. When g is credible and $S'(\tau) < 0$, delay can erode the option value of incumbent advantage.
5. **The correct strategy is staged validation.** Estimate (g, δ) with small experiments; commit when evidence supports high gain, acceptable drift, and time-sensitive niche capture.
6. **Experimentation has its own timing cost.** A pilot should be large enough to improve belief quality, but small and fast enough that it does not consume the niche window it is meant to evaluate.

13 Limitations and Empirical Calibration

This is still a stylized model. Skill is represented by a scalar log state, but real capability is multidimensional. The gain g may be an operator over capability vectors rather than a scalar. Drift δ should ideally be a distance in a capability space, possibly learned from data. Coordination cost c , niche timing $S(\tau)$, and the gain process may all be endogenous.

Empirical calibration would require proxies and an identification design. Gain g might be estimated from quality-adjusted output after adoption. Drift δ might be proxied by retraining time, temporary productivity loss, transfer-learning distance, task-embedding distance, or subjective friction. The coordination coefficient c might be proxied by organization size, management layers, cross-unit dependency, or deployment time. The niche coefficient $S(\tau)$ might be estimated from adoption curves, market share, standards lock-in, developer ecosystem growth, or first-mover retention.

A minimal empirical anchor would select one concrete switching setting and estimate a small number of primitives. For example, in an LLM-agent or HCI workflow, one could compare a baseline hand-coded or manual workflow with a new agentic workflow. A coarse calibration could measure pre-switch output speed, temporary productivity loss during retraining, post-switch quality-adjusted output speed, pilot duration, and subjective friction. These observations would not fully identify the general model, but they would provide an interpretable estimate of (g, δ) and test whether the model’s qualitative predictions match observed adoption behavior.

14 Conclusion

The revised model can be summarized as follows. Paradigm switching exchanges legacy log-skill for paradigm gain. Gain and drift are independent quantities. The core equation is

$$V^+ = e^{h(\delta)x^-} R^- g, \tag{83}$$

with decision statistic

$$\frac{V^+}{V^-} = ge^{(h(\delta)-1)x^-}. \quad (84)$$

A high-gain, low-drift switch is a powerful opportunity. A high-gain, high-drift switch can be strategically correct but initially painful. A low-gain, high-drift switch can erode value. Incumbents and high-legacy-skill experts may delay adoption because of real transfer losses. But when gain is verified and niches decay, delay may be the larger irrationality.

The log-skill version addresses the arbitrary scaling problem of direct K^h notation by anchoring skill to a novice baseline, integrates the ratio-based learning motivation, corrects the timing objective, preserves drift in repeated switching, and derives a minimal real-options characterization for staged validation under uncertain gain and drift. The final real-options extension separates perfect information from noisy validation and makes explicit that experimentation itself can erode niche advantage. The central strategic question is not whether to be conservative or innovative. It is whether the expected gain of a new paradigm is large enough, sufficiently transferable, and sufficiently verified to compensate for drift-induced compression of legacy skill and the timing cost of delay.

A Connection to the Earlier Scalar Formula

The earlier scalar model used

$$K^+ = K^{\min(r, 1/r)}, \quad R^+ = Rr. \quad (85)$$

For $r > 1$, this gives

$$\frac{V^+}{V^-} = rK^{1/r-1}. \quad (86)$$

The no-pain threshold is

$$rK^{1/r-1} \geq 1 \iff K \leq r^{r/(r-1)}. \quad (87)$$

The two-dimensional model generalizes this by replacing scalar gain with g and replacing the scalar transfer exponent $1/r$ with $h(\delta)$. Under the special identification $g = r$ and $h(\delta) = 1/r$, one recovers

$$\kappa^*(g, \delta) = g^{1/(1-h)} = r^{r/(r-1)}. \quad (88)$$

In the revised formulation, K should be read as dimensionless productivity multiple κ , or equivalently one should work in log-skill $x = \ln \kappa$.

B Asset Mapping

If total output is mapped to financial or real-world assets, define

$$O_{\text{asset}}(T) = mO(T), \quad (89)$$

where m is an environment or market-conversion multiplier. It may represent capital market depth, labor pricing, equity participation, global market access, institutional support, and other asset-conversion conditions. It should be interpreted as an environmental or institutional conversion factor, not as a personal trait.

References

- [1] Arrow, K. J. (1962). The economic implications of learning by doing. *Review of Economic Studies*, 29(3), 155–173.
- [2] Bass, F. M. (1969). A new product growth for model consumer durables. *Management Science*, 15(5), 215–227.
- [3] Bresnahan, T. F., and Trajtenberg, M. (1995). General purpose technologies: Engines of growth? *Journal of Econometrics*, 65(1), 83–108.
- [4] Christensen, C. M. (1997). *The Innovator's Dilemma*. Harvard Business School Press.
- [5] Cobb, C. W., and Douglas, P. H. (1928). A theory of production. *American Economic Review*, 18(1), 139–165.
- [6] Dixit, A. K., and Pindyck, R. S. (1994). *Investment Under Uncertainty*. Princeton University Press.
- [7] Klemperer, P. (1987). Markets with consumer switching costs. *Quarterly Journal of Economics*, 102(2), 375–394.
- [8] Kuhn, T. S. (1962). *The Structure of Scientific Revolutions*. University of Chicago Press.
- [9] March, J. G. (1991). Exploration and exploitation in organizational learning. *Organization Science*, 2(1), 71–87.
- [10] Nelson, R. R., and Winter, S. G. (1982). *An Evolutionary Theory of Economic Change*. Harvard University Press.
- [11] Rogers, E. M. (2003). *Diffusion of Innovations* (5th ed.). Free Press.
- [12] Teece, D. J. (2007). Explicating dynamic capabilities: The nature and microfoundations of enterprise performance. *Strategic Management Journal*, 28(13), 1319–1350.
- [13] Wright, T. P. (1936). Factors affecting the cost of airplanes. *Journal of the Aeronautical Sciences*, 3(4), 122–128.